$\mathbb{T}^2$-invariant Kähler-Einstein metrics on $\mathbb{CP}^2 \# 3\,\overline{\mathbb{CP}}^2$

A computationally assisted proof

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Einstein manifolds

Let (M, g) be a Riemannian manifold. It is *Einstein, if*

$$\mathrm{Ric}_g = \lambda g$$

where $\lambda = \{-1, 0, 1\}$. *In coordinates:*

$$\frac{1}{2} g^{st} \partial_{jt} g_{is} + \partial_{it} g_{js} - \partial_{st} g_{ij} - \partial_{ij} g_{st} + Q(g, \partial g) = \lambda g_{ij}$$

weakly elliptic, coupled, quasilinear system of $n(n+1)/2$ *equations in* n *variables.*

"The introduction of λ *was the biggest blunder of my life"*

- A. Einstein

Einstein metrics on M^4

To date, all known examples of Einstein metrics with $\lambda > 0$ on M^4 compact are obtained by imposing at least one of:

(a) Hermitian: (M, g, J), i.e. $g(J\cdot, J\cdot) = g(\cdot, \cdot)$ and/or,

(b) $\mathbb{T}^2 \curvearrowright g$ by isometries ($\mathbb{T}^2 := \mathbb{S}^1 \times \mathbb{S}^1$).

If Kähler ($N_J = 0$, $d\omega = 0$), the Einstein condition becomes:

$$\rho = \lambda\omega$$

where $\rho = \mathrm{Ric}_g(J\cdot, \cdot)$ is the Ricci form, $\omega = g(J\cdot, \cdot)$. Moreover,

$$[\rho] = 2\pi[c_1(M)] \in H^2_{dR}(M, \mathbb{R}).$$

Therefore if g is Einstein, $\lambda[\omega] = 2\pi[c_1(M)]$. Hence $c_1(M) > 0$ is a necessary condition for $\lambda > 0$ but not sufficient.

Existence & Uniqueness of g_E with $\lambda > 0$ on (M^4, J)

(D. Page, 1979)

Found a Hermitian Einstein metric on $(\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2, J, U(2))$

(G. Tian, 1990)

$\exists\ (M, g_{KE}, J) \iff \mathbb{CP}^2,\ \mathbb{CP}^1 \times \mathbb{CP}^1, \mathbb{CP}^2 \# k\,\overline{\mathbb{CP}}^2,\ 3 \leq k \leq 8$

(C. LeBrun et al., 2008)

Found a Kahler-Einstein metric on $(\mathbb{CP}^2 \# 2\,\overline{\mathbb{CP}}^2, J, \mathbb{T}^2)$

(Hitchin, 1974) $\implies \nexists(\mathbb{CP}^2 \# k\,\overline{\mathbb{CP}}^2, g_E),\ k > 8$

(T. Mabuchi et al., 1987),

$\implies$ Kähler-Einstein metrics are uniquely determined by (M, J).

(Corollary), the Kähler-Einstein metric on $\mathbb{CP}^2 \# 3\,\overline{\mathbb{CP}}^2 = Bl_{p_1 p_2 p_3}\mathbb{CP}^2$ with $J_{\text{blow-up}}$ is unique.

So why try find $(\mathbb{CP}^2 \# 3\,\overline{\mathbb{CP}}^2, \omega_{KE}, J)$ explicitly?

- $\mathbb{CP}^2, \mathbb{CP}^1 \times \mathbb{CP}^1, \mathbb{CP}^2 \# k\,\overline{\mathbb{CP}}^2$, $1 \leq k \leq 3$ are all Toric varieties, but not for $k \geq 4$,
- G. Tian, 1990 $\implies$ $k = 1, 2$ never Kähler (Futaki-invariant),
- $(\mathbb{CP}^2, \omega_{FS})$, $(\mathbb{CP}^1 \times \overline{\mathbb{CP}}^1, \omega_{FS} \oplus \omega_{FS})$ are explicitly known.

Only Kähler-Einstein toric metric with closed form not explicitly known

The program

How will we find the Einstein metric?

1. **Write** $\mathrm{Ric}_g = \lambda g$ in 'nice' coordinates.
2. **Solve** $\mathrm{Ric}_g = \lambda g$ numerically to high precision.
3. **Linearize** at the approximate metric that is close to being Einstein.
4. **Prove** a suitable map has a fixed point.

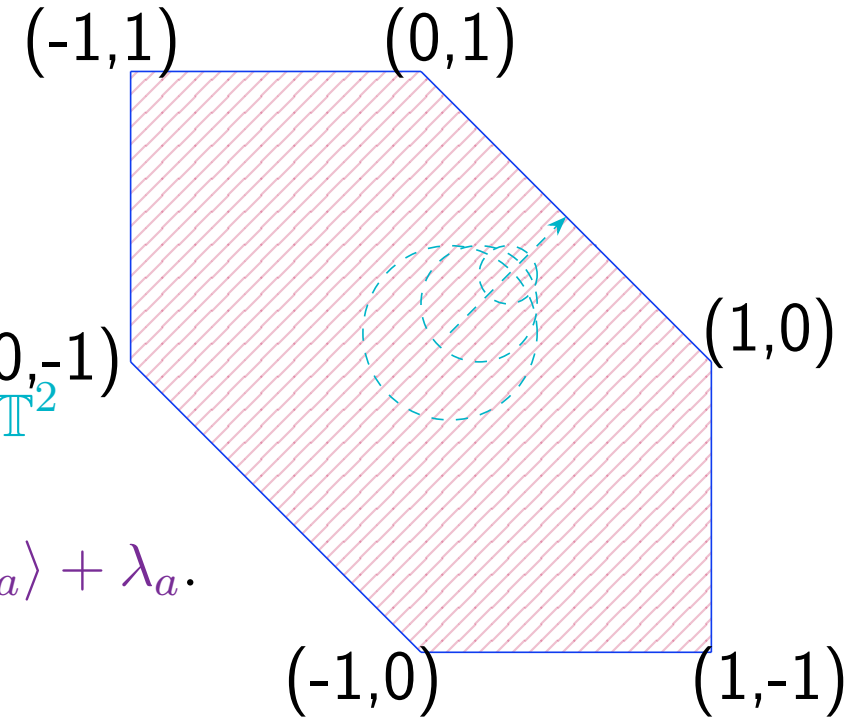
(Buttsworth, Hodgkinson, 24') found a new Einstein metric with $\lambda > 0$ on $(\mathbb{S}^{12}, O(3) \times O(10))$.

Step 1: Geometry & Topology on $M := \mathbb{CP}^2 \# 3\,\overline{\mathbb{CP}}^2$

$(\mathbb{CP}^2 \# 3\,\overline{\mathbb{CP}}^2, \omega_{KE}, J)$ admits an effective Hamiltonian action:

$\tau : \mathbb{T}^2 \to \mathrm{Diff}(\mathbb{CP}^2 \# 3\,\overline{\mathbb{CP}}^2, \omega, J)$

of $\mathbb{T}^2 := \mathbb{S}^1 \times \mathbb{S}^1$.



Key Point (1)

$M_0 := \{p \in M \mid \mathbb{T}^2 \curvearrowright p \text{ is free}\} \simeq P_0 \times \mathbb{T}^2$

where

$P = \{x \in \mathbb{R}^2 \mid l_a(x) \geq 0\},\ l_a(x) = \langle x, v_a \rangle + \lambda_a.$

Key Point (2):

(Abreu, 2000), $\quad g = u_{ij} dx_i dx_j + u^{ij} d\theta_i d\theta_j, u \in C^\infty(P_0)$

$$u_{ij} = \frac{1}{2} \sum_a \frac{v_a^i v_a^j}{l_a(x)} + h_{ij}(x), \qquad h \in C^\infty(P),\ \mathrm{Hess}(u) > 0 \text{ on } P_0.$$

Goal: Find h

Step 2: Approximate numerical solution

The Kähler-Einstein equation, $\rho(\tilde{\omega}) = \tilde{\omega}$ becomes an equation for ϕ,

$$(\omega + i\partial\bar{\partial}\phi)^2 = e^{F-\phi}\omega^2$$

where $F = \rho(\omega) - \omega$ such that $\tilde{\omega} = \omega + i\partial\bar{\partial}\phi$. Used a custom chebyshev solver for the *non-linear Monge–Ampère equation.*

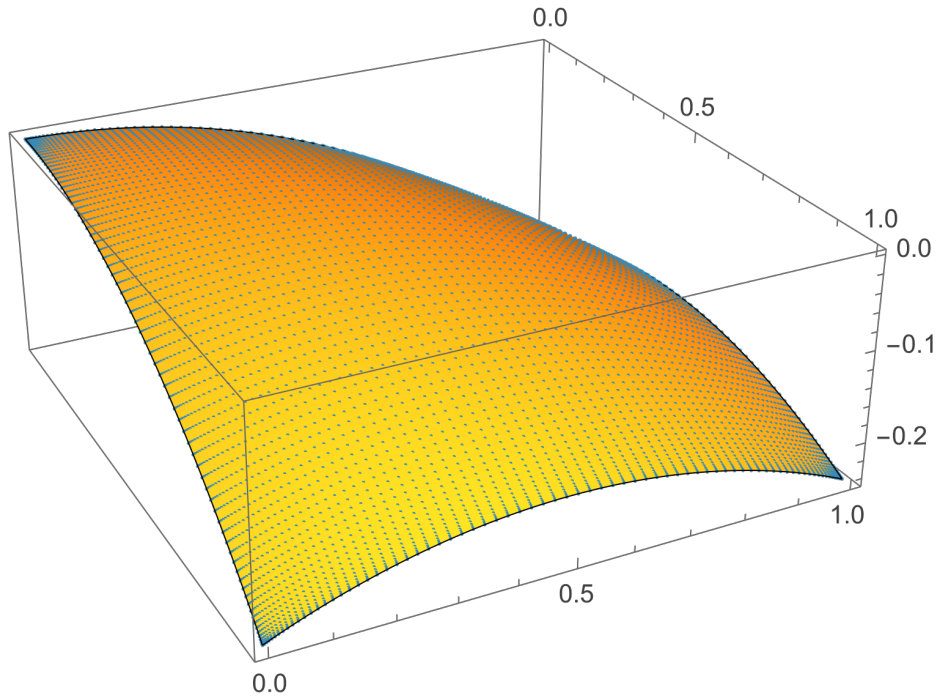


Figure: 80×80 degree Chebyshev polynomial solution on P/D_6 with grid points in blue.

Step 3: Linearization at $\tilde{g}$ and fixed point problem

The linearization of

$$(\omega + i\partial\bar{\partial}\phi)^2 = e^{F-\phi}\omega^2 \tag{1}$$

at 0 is

$$\mathscr{L}(\phi) = -\frac{1}{2}\Delta\phi + e^F\phi \tag{2}$$

where $\Delta := d^*d$.Hence we can write (1) as

$$0 = e + \mathscr{L}\phi + \mathscr{N}\phi. \tag{3}$$

Let $\xi = \mathscr{L}\phi$ and suppose $\mathscr{L}$ is invertible. Then (3) becomes

$$\xi = -e - \mathscr{N}\mathscr{L}^{-1}\xi \tag{4}$$

where

- $e := 1 - e^F$,
- $\mathscr{N}\phi := \frac{(i\partial\bar{\partial}\phi)^2}{\omega^2} - e^F(e^{-\phi} - 1 + \phi)$.

Note: h is $\mathbb{T}^2$-invariant so we care about spectral theory of **invariant eigenvalues**.

How do we prove $-e - \mathscr{N}\mathscr{L}^{-1}$ has a fixed point?

Step 4 - Estimates (Sketch)

$$\frac{1}{2} \star_\omega (i\partial\bar{\partial}\phi)^2 \text{ is quadratic, second order term.}$$

$$\implies \phi \in H^5(M).$$

Hence:

$$\mathscr{L} \in \mathcal{B}(H^5(M), H^3(M)) \ \& \ L^2(M) \text{ self–adjoint}$$

Key Estimate:

(Bochner Formula, $[\nabla, \Delta]$, integration by parts)

$$\implies \|\mathscr{L}\varphi\|_{H^3(M)} \geq C(\|\mathrm{Ric}\|_{C^0(M)}, \left\|\nabla^k \mathrm{Riem}\right\|_{C^0(M)}, \lambda_1) \|\varphi\|_{H^5(M)}$$

where $C > 0$ and $0 \leq k \leq 2$.

$$(\|e\|_{C^2} < \epsilon) \implies -e - \mathscr{N}\mathscr{L}^{-1} \text{ a contraction map on } B_0(\epsilon)$$

Note:

(a) $\lambda_1 > 0$,

(b) $\{\|\mathrm{Ric}\|_{C^0(M)}, \left\|\nabla^k \mathrm{Riem}\right\|_{C^0(M)}\} < \mathbf{K}$,

(c) $\|\mathrm{Ric}\|_{C^0(M)} > 0$.

Step 4 (b): Pointwise curvature estimates

The norm of the Riemannian curvature tensor with respect to the Riemannian metric g is quite simple.

$$\|\mathrm{Riem}\|_g^2 = u_{ab}^{ik} u_{ik}^{ab} \tag{5a}$$

$$\|\nabla \mathrm{Riem}\|_g^2 = u_{ikm}^{jl}(u^{mn}u_{jln}^{ik} + u_{nl}^{ik}u_{j}^{mn} - u_{jl}^{nk}u_{n}^{mi} + u_{jn}^{ik}u_{l}^{mn} - u_{jl}^{in}u_{n}^{mk}) \tag{5b}$$

where $u_{\alpha}^{ij} = \frac{\partial^{|\alpha|} u^{ij}}{\partial x^{\alpha_1} \cdots \partial x^{\alpha_k}}$.

Curvature norms are rational functions & $u^{ij} \in C^{\infty}(P)$

.

$$u^{ab} = \frac{\prod_c l_c}{Q} \mathrm{Adj}(u_{ij})_{ab}$$

With $Q > 0$ and $\|\prod_c l_c \mathrm{Adj}(u_{ij})\|_{C^k}(P) < \infty$.

Use Interval arithmetic to verify $\left\|\nabla^k \mathrm{Riem}\right\|_{C^0(M)}$ *a-posteori*

.

Bringing it all together

1. Used $\mathbb{T}^2 \curvearrowright (\mathbb{CP}^2 \# 3\,\overline{\mathbb{CP}}^2, \omega)$ to find nice coordinates.
2. Solved $\mathrm{Ric}_g = \lambda g$ numerically using chebyshev methods.
3. Formulated a fixed point problem in $H^3(\mathbb{CP}^2 \# 3\,\overline{\mathbb{CP}}^2)$ for existence.
4. Found suitable *a-posteori estimates using the numerical solution to prove existence of a "nearby" Einstein metric.*

Theorem (T. Buttsworth, W. Hadden, E. Hayes, D. Platt, W. Wiseman (Upcoming))

Let g_{KE} be the Kähler-Einstein metric on $\mathbb{CP}^2 \# 3\,\overline{\mathbb{CP}}^2$ (with respect to canonical complex structure) normalized to have volume $12\pi^2$. If g is the explicit approximate metric, then there is a diffeomorphism $\Phi : M \to M$ so that $\|g - \Phi^* g_{KE}\|_{H^3_g(M)} \leq 10^{-20}$.

Application: HSC and Rational Connectedness (RC)

Let (X, ω, J) be a compact Kähler manifold.

(S-T Yau, 1982, Yang 2018), $\mathrm{HSC}_\omega > 0 \implies X$ projective & RC.

Suppose $(E, \bar{\partial}_E, h) \to (X, J)$ is a Hermitian holomorphic vector bundle.

$(E, \bar{\partial}_E, h)$ is *uniformly* RC-positive if:

$\forall q \in X, \exists u \in T_q X$ such that $\forall v \in E_q \neq 0$,

$R^E(u, \overline{u}, v, \overline{v}) > 0.$

Suppose X projective. Conjecture (X. Yang, 2021):

X rationally connected $\iff$ $T^{1,0}X$ is uniformly RC-positive

(X. Yang, 2021) has shown "$\Longleftarrow$" what about the converse?

Corollary (Upcoming)

The holomorphic sectional curvature of the Kähler-Einstein metric g_{KE} on $(\mathbb{CP}^2 \# 3\,\overline{\mathbb{CP}}^2, J)$ changes sign.